

AQI-Predictor: End-to-End Air-Quality Forecasting for Lahore

Technical Project Report | Hopworks, XGBoost, FastAPI, Streamlit, and GitHub Actions

Abstract / Executive Summary

AQI-Predictor forecasts Lahore’s Air Quality Index (AQI) up to three days ahead using pollution history, OpenWeather forecasts, and engineered temporal features. Hopworks manages features and models, GitHub Actions automates pipelines, FastAPI serves predictions, and Streamlit presents results. XGBoost was the strongest of four tested algorithms, achieving test $R^2 = 0.9875$, $MAE = 1.8053$, and $RMSE = 3.7422$.

System Architecture & Data Pipeline

The feature pipeline collects CO, NO₂, O₃, PM_{2.5}, PM₁₀, temperature, and humidity from OpenWeather. Missing values use chronological interpolation, outliers are capped at the 99th percentile, and continuous variables are standardized. The scaler is versioned with the model.

The correlation matrix shows a 0.97 relationship between PM_{2.5} and PM₁₀, so PM₁₀ was removed to limit multicollinearity and leakage risk. Lagged AQI, rolling averages, change rate, and calendar fields capture temporal behavior.

Hopworks reports 64 features across four feature groups. In `aqi_features` version 4, `city` is the primary key and `timestamp` is the event-time field, providing a versioned schema for training and inference.

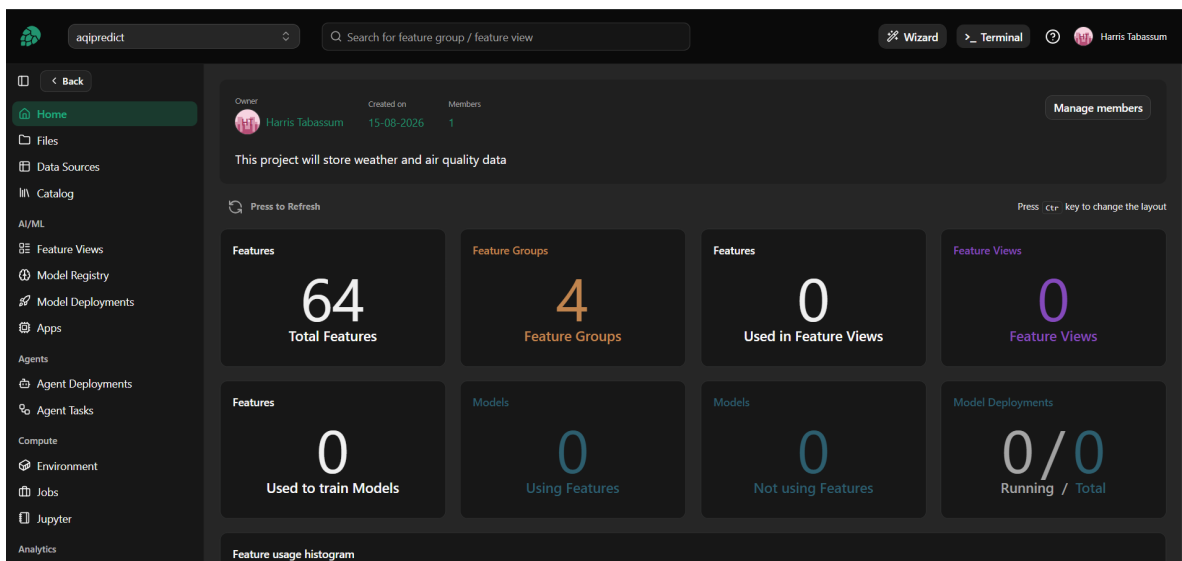


Figure 1. Hopworks project overview and feature-store inventory.

aqi_features 4 (latest)

Feature

Find feature

Primary Keys Partition Keys Foreign Keys Event Time Feature On-Demand

name	type	description
city	string	No description
timestamp	timestamp	No description
aqi	bigint	No description
co	double	No description
no2	double	No description
o3	double	No description
pm2_5	double	No description

Show all

Figure 2. Versioned feature schema in the `aqi_features` feature group.

aqi_predictor_v3 2 (latest) all versions --

Version created on 2026-08-22 20:27:14 Framework PYTHON

Best performing model: XGBoost

This version is not used in any deployments

Model metrics

MAE	R2	RMSE
1.0206449031829834	0.991931438460449	3.009014368057251

Figure 3. Model Registry entry for `aqi_predictor_v3`.

CI/CD & Pipeline Automation

GitHub Actions runs scheduled workflows for daily feature ingestion, model retraining, and inference setup. New observations are validated before entering Hopsworks; training jobs compare candidate models and register the winner with its scaler. At startup, the API retrieves the latest approved artifacts. Repository secrets protect service credentials and repeatable jobs reduce manual deployment work.

Modeling Strategy & Results

Ridge Regression, Random Forest, XGBoost, and a TensorFlow RNN were evaluated using chronological train and test sets. Their complete results are summarized below.

Model	MAE		RMSE		R^2	
	Train	Test	Train	Test	Train	Test
Ridge	6.68	7.19	10.10	10.29	0.9900	0.9057
Random Forest	3.83	4.71	8.30	7.16	0.9932	0.9543
XGBoost	2.00	1.81	4.31	3.74	0.9982	0.9875
RNN	8.61	10.39	12.02	13.65	0.9858	0.8339

Table 1. Model evaluation results on chronological train and test sets.

Best Model: XGBoost (Test $R^2 = 0.9875$).



Figure 4. Comparative model leaderboard in the evaluation tab.

SHAP ranks $PM_{2.5}$, lagged AQI, and CO as the leading drivers, while residuals remain concentrated near zero. The booster is stored as native XGBoost `.json` to avoid cross-platform pickle problems. Hopsworks lists `aqi_predictor_v3` version 2 with $R^2 = 0.9919$, MAE = 1.0206, and RMSE = 3.0090.

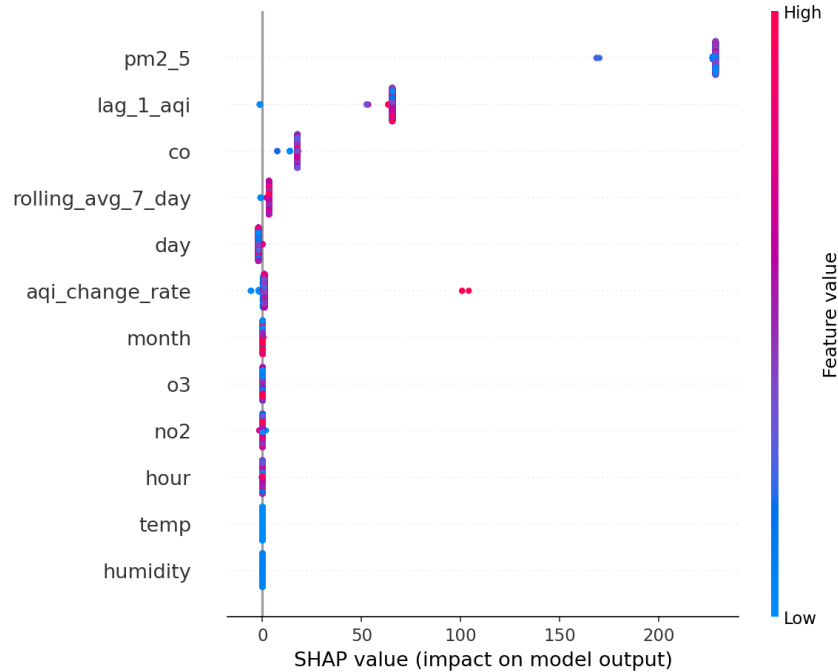


Figure 5. SHAP feature impacts on model output.

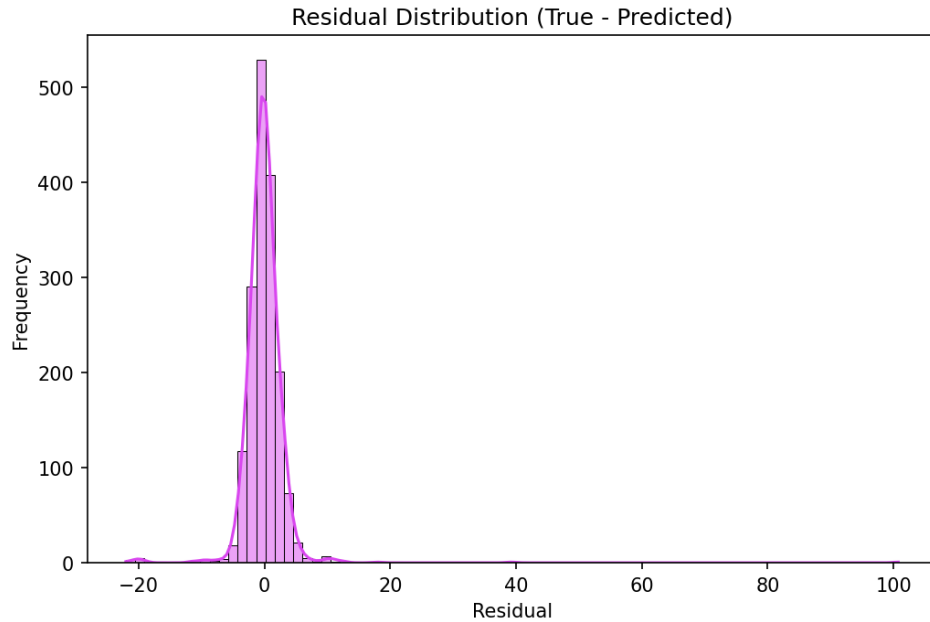


Figure 6. Residual distribution (true minus predicted).

Dashboard & Deployment

FastAPI downloads the latest model and scaler from Hopsworks, retrieves the live three-day weather outlook, transforms the inputs, and returns AQI predictions. The supplied images show the dashboard rather than API documentation, so endpoint names are not inferred.

Streamlit provides *Forecast & Alerts*, *EDA*, and *Model Evaluation* tabs. The screenshot reports AQI 93 (“Moderate”) with pollutant details. Forecast cards show 87, 118, and 113, and the Plotly trend highlights the rise into “Unhealthy for Sensitive Groups.” AQI above 300 triggers a hazardous warning.

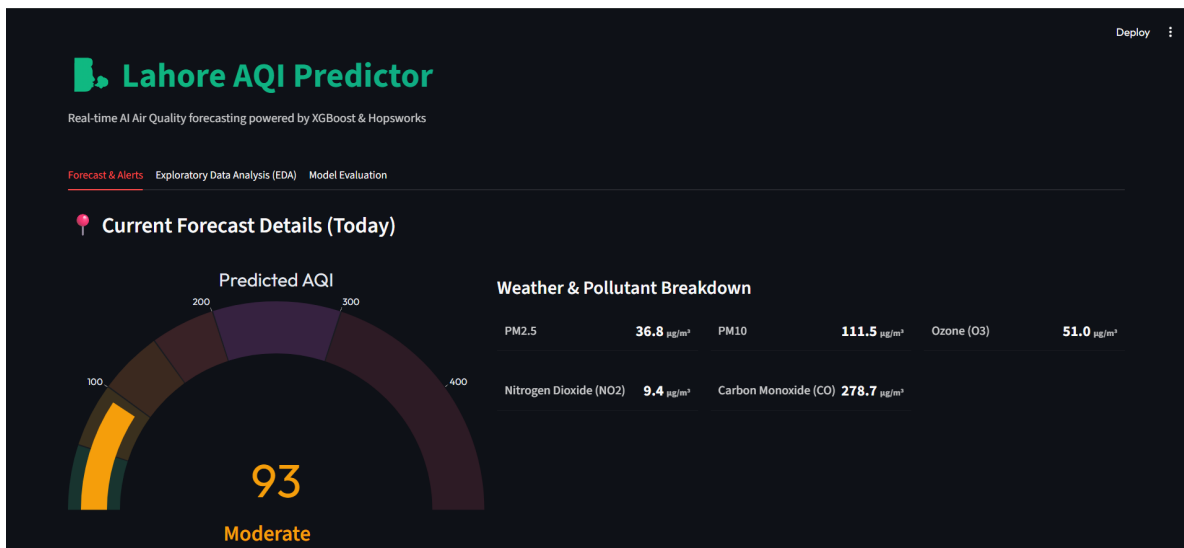


Figure 7. Current AQI gauge and pollutant breakdown.

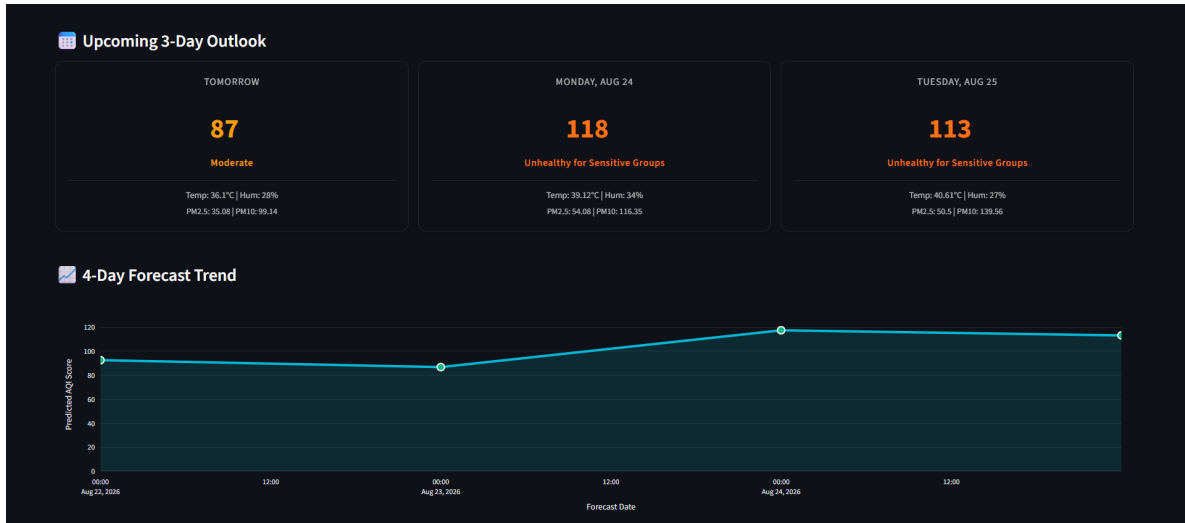


Figure 8. Three-day outlook and four-day forecast trend.

The EDA tab shows the $PM_{2.5}$ –AQI relationship and correlation matrix; the evaluation tab presents the leaderboard, SHAP summary, and residuals.

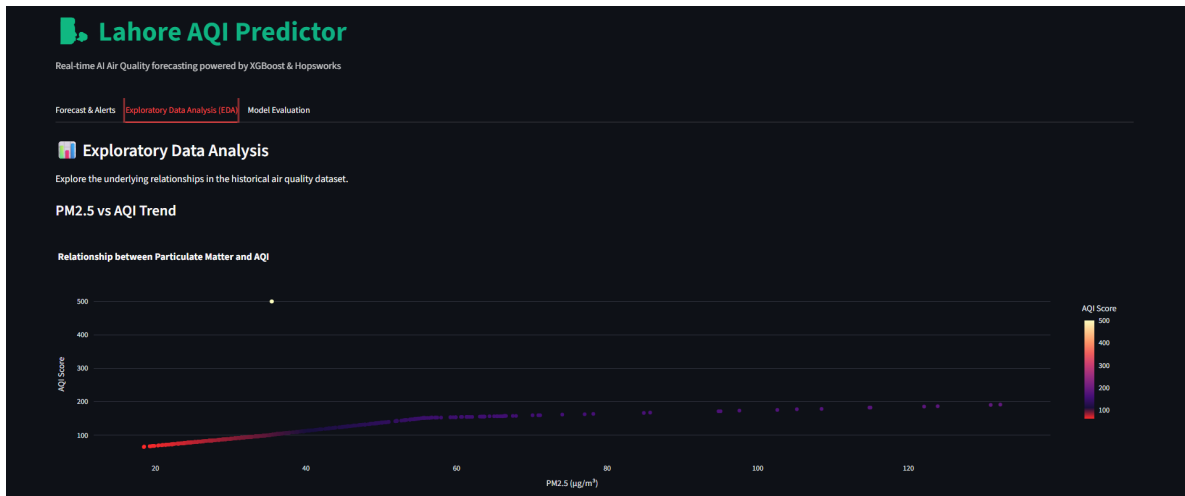


Figure 9. $PM_{2.5}$ –AQI scatter relationship.

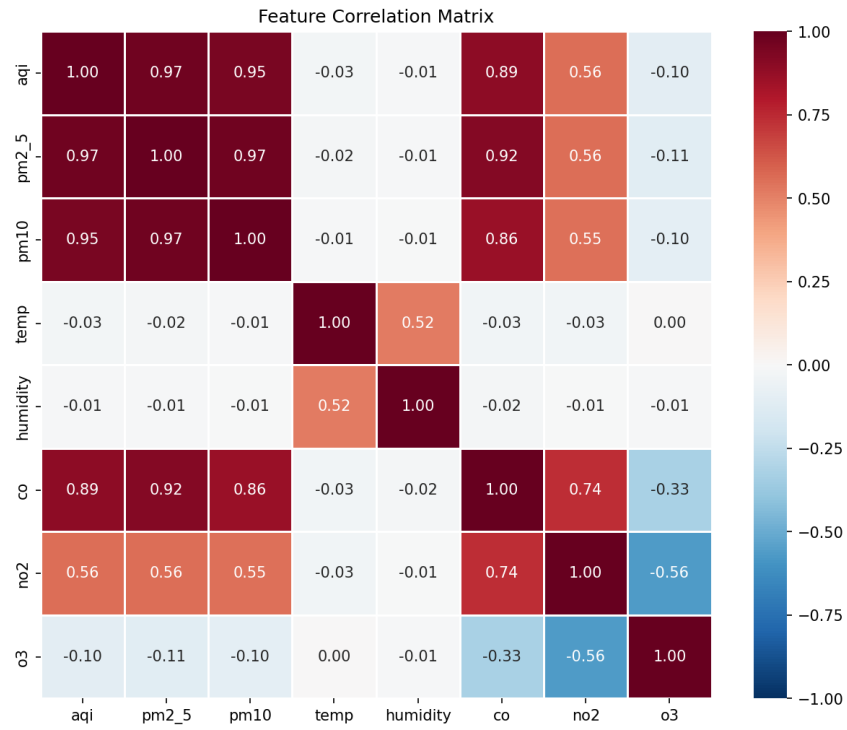


Figure 10. Feature correlation matrix.

Conclusion

AQI-Predictor combines governed data, automated pipelines, accurate XGBoost forecasting, explainability, and web deployment. Future work should add drift monitoring, prediction intervals, API health checks, and validation during extreme pollution events.